

A BLOCK-NORM COUNTEREXAMPLE IN THE $L^p(L^q)$ DILATION PROBLEM

AGENT LANE 04

STATUS

This packet gives a candidate counterexample to the equality question in Open Problem 7.6 of Fackler–Gluck [1]. Let

$$X_{p,q} = L^p([0, 1]; L^q([0, 1])), \quad 1 < p, q < \infty,$$

and let $\mathcal{T}_{p,q}$ be the weak operator closed convex hull of the positive invertible isometries of $X_{p,q}$. The packet proves, using the standard Guerre–Raynaud fibre form for positive isometries of $L^p(L^q)$, that

$$\mathcal{T}_{p,q} \neq \{T \in \mathcal{L}(X_{p,q}) : T \geq 0, \|T\| \leq 1\} \quad (p \neq q).$$

For $p = q$, Fubini identifies $X_{p,p}$ with scalar $L^p([0, 1]^2)$, and the scalar result used in the source gives equality. Thus the first question in Open Problem 7.6 has the sharp yes/no answer: equality holds for $p = q$ and fails for $p \neq q$. The characterization question for $p \neq q$ remains open.

SOURCE OPEN PROBLEM

The source asks:

Let $p, q \in (1, \infty)$ and let $\mathcal{T}$ denote the weak operator closure of the convex hull of all positive invertible isometries on $L^p([0, 1]; L^q([0, 1]))$. Does $\mathcal{T}$ coincide with the set of all positive contractions? If not, can a good characterization of the elements of $\mathcal{T}$ be given?

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Open Problem 7.6. Let $p, q \in (1, \infty)$ and let $\mathcal{T}$ denote the weak operator closure of the convex hull of all positive invertible isometries on $L^p([0, 1]; L^q([0, 1]))$. Does $\mathcal{T}$ coincide with the set of all positive contractions? If not, can a good characterization of the elements of $\mathcal{T}$ be given?

Dilations on Hilbert spaces. In the previous subsection we considered a dilation

IDEA OF THE PROOF

Operators in $\mathcal{T}_{p,q}$ inherit a finite-dimensional shadow from the fibre form of positive invertible isometries. Compress such an operator to a rectangular step sublattice

$$E = \ell_p^2(\ell_q^2) \subset L^p([0, 1]; L^q([0, 1])).$$

If $R \in \mathcal{T}_{p,q}$ and $R_{ij} : \ell_q^2 \rightarrow \ell_p^2$ are the four outer blocks of this compression, then the scalar matrix

$$(\|R_{ij}\|)_{i,j=1}^2$$

must be dominated by a positive contraction on ℓ_p^2 . The reason is that positive invertible isometries act as an outer scalar L^p isometry together with fibrewise positive L^q isometries; the outer

scalar compression supplies the dominating ℓ_p contraction, and convexity plus weak closure preserve the domination.

We then write down a positive contraction on $\ell_p^2(\ell_q^2)$ whose block norm matrix is either

$$\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \quad (q > p), \quad \text{or} \quad \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \quad (p > q).$$

These matrices have ℓ_p^2 operator norm $2^{1/p'}$ and $2^{1/p}$, respectively, both strictly larger than one. The lifted positive contraction therefore cannot belong to $\mathcal{T}_{p,q}$.

THE FINITE COMPRESSION OBSTRUCTION

Let A_1, A_2 be a measurable partition of $[0, 1]$ into two sets of positive measure, and let B_1, B_2 be another such partition. Let $E = \ell_p^2(\ell_q^2)$, with vectors written

$$x = ((x_{11}, x_{12}), (x_{21}, x_{22})).$$

Embed E isometrically into $X_{p,q}$ as the rectangular step sublattice by using normalized indicators of the four rectangles $A_i \times B_\alpha$. Denote this embedding by $J : E \rightarrow X_{p,q}$. Let P be conditional expectation onto this rectangular sigma algebra and set $Q = J^{-1}P$. Then J is a positive isometry, Q is a positive contraction, and $QJ = I_E$.

Lemma 1 (block domination). *Assume $p \neq q$. If $R \in \mathcal{T}_{p,q}$ and*

$$QRJ = (R_{ij})_{i,j=1}^2$$

is written as a 2×2 matrix of operators $R_{ij} \in \mathcal{L}(\ell_q^2)$, then the scalar matrix $B_R = (\|R_{ij}\|_{i,j=1}^2)$ satisfies

$$\|B_R\|_{\ell_p^2 \rightarrow \ell_p^2} \leq 1.$$

Proof. It suffices first to prove the assertion for a positive invertible isometry U of $X_{p,q}$. We use the standard Guerre–Raynaud form for positive isometries of $L^p(L^q)$, in the form quoted for $L_p(L_q)$ in [2, proof of Proposition 6.12]: a positive invertible isometry acts by a positive scalar L^p isometry on the outer variable and by a measurable field of positive L^q isometries on the fibres. Equivalently, there are a positive invertible isometry S of scalar $L^p[0, 1]$ and fibre isometries V_s such that the outer norm function of Uf is the scalar isometric image under S of the outer norm function of f .

Let $J_p : \ell_p^2 \rightarrow L^p[0, 1]$ be the normalized two-step embedding associated with A_1, A_2 , and let $Q_p : L^p[0, 1] \rightarrow \ell_p^2$ be the corresponding conditional expectation quotient. The scalar compression

$$C = Q_p S J_p$$

is a positive contraction on ℓ_p^2 ; write $C = (c_{ij})_{i,j=1}^2$. For $x \in \ell_q^2$ placed in the j th outer coordinate of E , the fibre isometries do not increase the subsequent conditional expectation onto $\text{span}\{\chi_{B_1}, \chi_{B_2}\}$. Hence the i, j block of QUJ has norm at most c_{ij} :

$$\|(QUJ)_{ij}x\|_{\ell_q^2} \leq c_{ij}\|x\|_{\ell_q^2}.$$

Thus the block norm matrix of QUJ is dominated entrywise by the positive ℓ_p^2 contraction C .

Now let R be in the convex hull of positive invertible isometries. The block norm matrix of QRJ is entrywise dominated by the same convex combination of the corresponding scalar compression matrices, hence by a positive ℓ_p^2 contraction. Finally, if $R_\alpha \rightarrow R$ weakly in $\mathcal{L}(X_{p,q})$, then $QR_\alpha J \rightarrow QRJ$ weakly on the finite-dimensional space E , hence in norm. The block norms therefore converge, so the conclusion passes to the weak operator closure. $\square$

THE COUNTEREXAMPLE

Theorem 1. *Let $1 < p, q < \infty$ and $p \neq q$. Then $\mathcal{T}_{p,q}$ is a proper subset of the positive contractions on $X_{p,q}$.*

Proof. We construct the finite model first, then lift it through J and Q .

Suppose first that $q > p$. Define $S_> : \ell_p^2(\ell_q^2) \rightarrow \ell_p^2(\ell_q^2)$ by

$$S_>((a, b), (c, d)) = ((a, d), (0, 0)).$$

This is positive. Moreover

$$\|S_>((a, b), (c, d))\| = (|a|^q + |d|^q)^{1/q} \leq (\|(a, b)\|_q^p + \|(c, d)\|_q^p)^{1/p},$$

because $q > p$ implies $\|\cdot\|_{\ell_q^2} \leq \|\cdot\|_{\ell_p^2}$ on non-negative two-vectors. Hence $S_>$ is a positive contraction. Its outer blocks have norm matrix

$$B_{S_>} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix},$$

whose norm on ℓ_p^2 is $2^{1/p'} > 1$.

Suppose next that $p > q$. Define $S_< : \ell_p^2(\ell_q^2) \rightarrow \ell_p^2(\ell_q^2)$ by

$$S_<((a, b), (c, d)) = ((a, 0), (0, b)).$$

Again $S_<$ is positive, and

$$\|S_<((a, b), (c, d))\| = (|a|^p + |b|^p)^{1/p} \leq (|a|^q + |b|^q)^{1/q} \leq (\|(a, b)\|_q^p + \|(c, d)\|_q^p)^{1/p},$$

so $S_<$ is a positive contraction. Its block norm matrix is

$$B_{S_<} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix},$$

whose norm on ℓ_p^2 is $2^{1/p} > 1$.

In either case call the relevant finite contraction S . Set

$$T = JSQ \in \mathcal{L}(X_{p,q}).$$

Since J is positive isometric, Q is a positive contraction, and S is a positive contraction, T is a positive contraction on $X_{p,q}$. But

$$QTJ = Q(JSQ)J = S.$$

If T belonged to $\mathcal{T}_{p,q}$, Lemma 1 would imply that the block norm matrix of S has ℓ_p^2 norm at most one. The explicit matrices above have norms $2^{1/p'}$ or $2^{1/p}$, a contradiction. Thus $T \notin \mathcal{T}_{p,q}$. $\square$

CONSEQUENCES AND LIMITATIONS

For $p = q$, Fubini identifies $X_{p,p}$ with scalar $L^p([0, 1]^2)$, and the source recalls the scalar Greim theorem that positive contractions on atomless scalar L^p are weak operator limits of convex combinations of positive invertible isometries. Thus equality holds in the diagonal case $p = q$. Theorem 1 shows equality fails off the diagonal.

This does not provide a full characterization of $\mathcal{T}_{p,q}$ when $p \neq q$. The obstruction suggests that such a characterization must include outer-block domination constraints, not merely positivity and contractivity.

NOVELTY AND VERIFICATION NOTES

- Local indexes were searched for 1709.08547, $L^p(L^q)$, positive invertible isometries, and counterexample phrases. The only exact prior packet was the earlier tensor-product partial result for the same source problem.
- Bounded web search on 2026-06-29 for the source title, Open Problem 7.6, positive invertible isometries on $L^p(L^q)$, and weak operator closure found no later paper giving this counterexample or a full characterization.
- The finite norm checks are explicit: the two displayed block matrices have norms $2^{1/p'}$ and $2^{1/p}$ on ℓ_p^2 .
- The main human-review point is Lemma 1, specifically the use of the standard Guerre–Raynaud fibre representation for positive invertible isometries of $L^p(L^q)$ throughout the stated exponent range $p \neq q$.

HUMAN REVIEW RECOMMENDATION

Review as a likely valid counterexample and negative answer to the equality part of Open Problem 7.6. If the Guerre–Raynaud fibre form is accepted in the stated positive-isometry range, the rest of the proof is a finite two-by-two norm computation.

REFERENCES

- [1] S. Fackler and J. Gluck, *A Toolkit for Constructing Dilations on Banach Spaces*, arXiv:1709.08547, 2017. <https://arxiv.org/abs/1709.08547>
- [2] M. Cúth, N. de Rancourt, and M. Doucha, *Guarded Fraïssé Banach spaces*, arXiv:2408.12755, 2024. <https://arxiv.org/abs/2408.12755>
- [3] S. Guerre and Y. Raynaud, *Sur les isometries de $L^p(X)$ et le theoreme ergodique vectoriel*, Canad. J. Math. 40 (1988), no. 2, 360–391.